SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: CRYPTOGRAPHY AND NETWORK SECURITY
COURSE CODE: CSA51

COURSE OUTCOME COVERED

CO2: Analyze Public Key crypto system strategies using RSA and key Exchange for Encryption algorithm to strengthen information security. (BL4,BL5)

Scenario-Based : 20%

Total Marks: 50

Annexure A

Code of Conduct:

I A. Midhuna (192521302) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

A. Midhuna
Signature of the student:

Annexure A

1. Secure Password Storage in Web Applications

A web application stores user passwords and needs to protect them against database breaches.

A. Explain why storing plaintext passwords is insecure.

Scenario: A company develops a web application where users create accounts using their email addresses and passwords. The application stores all passwords in plaintext in its database.

If hackers gain access to the database, they can directly read every user's password. They may use these passwords to log in to users' accounts or access other websites where users have reused the same passwords.

Problems:

1. Passwords can be easily viewed if the database is leaked.
2. Hackers can misuse users' accounts and personal information.
3. Reused passwords can expose multiple accounts.
4. The company may lose customer trust and face security consequences.

B. Suggest a secure method for storing passwords and justify your choice.

Solution: The company should use Argon2id password hashing with a unique salt for every password.

Working process:

1. The user enters a password during registration.
2. The application generates a unique random salt.
3. Argon2id combines the password and salt to generate a secure hash.
4. The application stores the hash and salt in the database instead of the original password.
5. During login, the application hashes the entered password using the stored salt and verifies the result.

Advantages:

- One-way protection: The original password cannot feasibly be recovered from the hash.
- Unique salt: Prevents identical passwords from producing identical stored hashes when different salts are used.
- Brute-force resistance: Argon2id is designed to make repeated password guesses computationally expensive.
- Better security: Even if attackers steal the database, they cannot immediately read the original passwords.

Conclusion

The company should never store passwords in plaintext. Using Argon2id with unique salts and appropriately configured parameters provides strong protection against password theft and offline guessing attacks. This helps protect user accounts even if the database is compromised.

2. Digital Signature for Document Authentication
A company wants to ensure that official documents sent electronically are authentic and unaltered.

A. Explain how digital signatures can be used in this scenario.

Scenario: A company sends official documents, such as contracts, invoices, and agreements, electronically to clients. The company needs to ensure that the documents are genuine and have not been modified during transmission.

Solution: The company can use a Digital Signature based on public-key cryptography.

Working process:

1. The sender creates an electronic document.
2. A hash value is generated from the document.
3. The sender signs the hash using their private key.
4. The signed document is sent to the receiver.
5. The receiver uses the sender's public key to verify the digital signature and checks the document's hash.

If the verification is successful, the receiver can confirm that the document has not been altered and that the signature corresponds to the sender's private key.

B. Justify how this method ensures both integrity and non-repudiation.

1. Integrity:

- A hash value is generated from the document's contents.
- If anyone modifies the document, its hash value changes.
- Signature verification will fail if the document has been altered.

Therefore, digital signatures help ensure that the document remains unchanged after signing.

2. Non-repudiation:

- The sender signs the document using their private key.
- The corresponding public key is used to verify the signature.
- If the private key is securely controlled and the identity is properly verified, the sender has strong evidence that they signed the document and cannot easily deny it later.

3. Additional points for B:

- Digital signatures use public-key cryptography to verify the authenticity of the sender.
- A digital certificate issued by a trusted Certificate Authority (CA) can help confirm the sender's identity.
- Thus, digital signatures provide reliable evidence of document authenticity and help prevent unauthorized modifications or false denial of signing.

Conclusion

Digital signatures provide document integrity, sender authentication, and support for non-repudiation. They help companies exchange official documents securely and reliably over electronic networks.

RUBRICS FOR CALCULATION:

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Misinterprets or fails to understand the scenario
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts
Decision Making	20%	Makes well-reasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand